

✓ Welcome

I have been hired by a 40-year-old news company called FF-NEWS. They have provided me with a list of their news headlines from March 2004. They are seeking 10 headlines from their newspaper specifically related to `climate change and global warming`.

As an AI Engineer, my task was to utilize the OpenAI text embedding model to identify 10 headlines about climate change and global warming from their archive.

```
!pip install openai python-dotenv

import pandas as pd
import os
from openai import OpenAI
from dotenv import load_dotenv
import plotly.express as px
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity

# Loading API key and organization ID from a dotenv file
load_dotenv(dotenv_path='apikey.env.txt')

# Retrieving API key and organization ID from environment variables
APIKEY = os.getenv("APIKEY")
ORGID = os.getenv("ORGID")

# Creating an instance of the OpenAI client with the provided API key and organization ID
client = OpenAI(
    # organization= ORGID,
    api_key=APIKEY
)
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client
```

Requirement already satisfied: openai in /usr/local/lib/python3.12/dist-packages (2.32.0)

Requirement already satisfied: python-dotenv in /usr/local/lib/python3.12/dist-packages (1.2.2)

```

Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from openai) (4.13.0)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from openai) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from openai) (0.28.1)
Requirement already satisfied: jiter<1,>=0.10.0 in /usr/local/lib/python3.12/dist-packages (from openai) (0.14.0)
Requirement already satisfied: pydantic<3,>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from openai) (2.12.3)
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from openai) (1.3.1)
Requirement already satisfied: tqdm>4 in /usr/local/lib/python3.12/dist-packages (from openai) (4.67.3)
Requirement already satisfied: typing-extensions<5,>=4.11 in /usr/local/lib/python3.12/dist-packages (from openai) (4.15.0)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5,>=3.5.0->openai) (3.13)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->openai) (2026.4.22)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->openai) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->openai) (0.14.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->openai) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->openai) (2.41.4)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->openai) (0.9.0)
<openai.OpenAI at 0x7eef2bd24a10>

```

Here is the list of news headlines

```

headlines = [
    "Political leaders convene for summit on climate change",
    "New breakthrough in renewable energy technology announced",
    "Global warming activists rally in major cities worldwide",
    "Sports teams unite to raise awareness about climate crisis",
    "Government announces new policy to combat climate change",
    "Tech giant unveils revolutionary AI-powered gadget",
    "Climate scientists warn of irreversible damage to the planet",
    "Political turmoil grips nation as elections approach",
    "Athletes advocate for environmental conservation efforts",
    "Tech startups compete for funding in Silicon Valley",
    "Rising temperatures threaten biodiversity, experts warn",
    "Sports championship marred by controversy over doping allegations",
    "New legislation aims to reduce carbon emissions",
    "Tech company announces plans for expansion into new markets",
    "Global warming effects felt in Arctic region, scientists say",
    "Pi Disk: 20.49 GB/107.72 GB sh over proposed climate change regulations",
    "Team achieves victory in sporting event despite odds",
    "Breakthrough in renewable energy research promises brighter future",
    "Climate summit ends with pledges for carbon neutrality",
    "Athlete breaks world record in thrilling sporting event",
    "Tech industry faces criticism over data privacy concerns",

```

"Government launches initiative to promote green technology",
"Climate change impacts highlighted in new scientific report",
"Political scandal rocks nation's capital",
"Sports star announces retirement after illustrious career",
"Tech company introduces innovative solution to environmental challenges",
"Global warming exacerbates natural disasters, experts warn",
"Political candidates debate strategies to address climate crisis",
"Sports league adopts sustainability measures to reduce carbon footprint",
"Tech expo showcases latest advancements in artificial intelligence",
"Climate activists stage protest outside government buildings",
"Political leaders vow to prioritize climate change action",
"Athlete overcomes injury to win gold medal",
"Tech company partners with environmental organizations for conservation projects",
"Global warming threatens food security, scientists caution",
"Political campaign focuses on environmental policies",
"Sports event canceled due to extreme weather conditions",
"Tech startup revolutionizes transportation industry with electric vehicles",
"Climate change summit attracts attention from around the world",
"Political unrest leads to protests in major cities",
"Sports team celebrates victory with championship parade",
"Tech industry grapples with cybersecurity challenges",
"Global warming impacts discussed at international conference",
"Political leaders face scrutiny over handling of climate crisis",
"Athlete honored with prestigious award for sportsmanship",
"Tech conference explores the future of artificial intelligence",
"Climate activists demand immediate action from world leaders",
"Political debate intensifies ahead of election day",
"Sports fans rally behind team in championship match",
"Tech company accused of monopolistic practices",
"Global warming solutions proposed at climate change forum",
"Political summit focuses on diplomatic relations",
"Athlete achieves personal best in sporting competition",
"Tech industry leaders testify before congressional committee",
"Climate change effects seen in rising sea levels, researchers find",
"Political parties clash over environmental policies",
"Sports league implements measures to promote diversity and inclusion",
"Tech company secures \$100M in funding for groundbreaking project",
"Global warming awareness campaign gains momentum",
"Political leaders spar over economic policies",
"Athlete inspires youth through community outreach programs",
"Tech company launches new product to improve quality of life",

"Climate change activists call for divestment from fossil fuels",
"Political commentator discusses implications of recent events",
"Sports tournament draws record-breaking viewership",
"Tech industry grapples with ethical dilemmas of AI",
"Global warming impact on wildlife habitats documented in new study",
"Political rally attracts thousands of supporters",
"Athlete makes comeback after overcoming adversity",
"Tech conference showcases cutting-edge innovations",
"Climate change legislation faces opposition in parliament",
"Political upheaval leads to government reshuffle",
"Sports team embarks on goodwill tour to promote peace",
"Tech company releases annual sustainability report",
"Global warming awareness raised through art and music festival",
"Political campaign enters final stretch with heated debates",
"Athlete named ambassador for youth sports program",
"Tech industry leaders meet to discuss future trends",
"Climate change protesters disrupt international summit",
"Political scandal unfolds with leaked documents",
"Sports star launches foundation to support underprivileged youth",
"Tech startup awarded for innovation in environmental sustainability",
"Global warming impact on agriculture highlighted in report",
"Political leaders negotiate international trade agreements",
"Athlete honored with induction into Hall of Fame",
"Tech company invests in renewable energy research",
"Climate change task force formed to address urgent issues",
"Political tensions escalate in region, raising concerns",
"Sports organization partners with charity for humanitarian efforts",
"Tech expo features demonstrations of virtual reality technology",
"Global warming debate intensifies with new scientific findings",
"Political campaign focuses on grassroots activism",
"Athlete advocates for gender equality in sports",
"Tech industry pioneers explore potential of blockchain technology",
"Climate change summit results in historic agreement",
"Political leaders reach compromise on controversial legislation",
"Sports team wins championship with dramatic final play",
"Tech company launches initiative to bridge digital divide",
"Global warming impact on agriculture highlighted in report",
"Political candidates engage voters in town hall meetings",
"Athlete inspires next generation through mentorship program",
"Tech conference showcases breakthroughs in quantum computing",
"Climate change action plan receives bipartisan support",

```
"Political movement gains momentum with widespread support",
"Sports star honored with prestigious sportsmanship award",
"Tech industry leaders advocate for diversity and inclusion initiatives",
"Global warming impact on coastal communities examined in documentary",
"Political summit addresses refugee crisis and humanitarian aid",
"Athlete donates winnings to charity for children's education",
"Tech startup disrupts industry with innovative business model",
"Climate change awareness campaign launches on social media",
"Political leaders engage in diplomatic talks to promote peace",
"Sports league implements strict anti-doping measures",
"Tech company pledges to reduce carbon footprint with sustainability initiatives",
"Global warming research expedition uncovers new insights",
"Political unrest sparks protests and civil unrest",
"Athlete breaks barriers as first in their sport",
"Tech industry leaders collaborate on open-source projects",
"Climate change documentary wins prestigious film award",
"Political candidates make final push in campaign rallies",
"Sports team celebrates victory with parade through city streets",
"Tech conference addresses cybersecurity threats and solutions",
"Global warming impact on indigenous communities addressed at UN summit"
]
```

```
import pandas as pd

# 1. Convert your original list of headlines into a DataFrame column
headlines_df = pd.DataFrame({"headline_text": headlines})

# 2. Extract the embeddings (your existing working line)
vector_embeddings = [d.embedding for d in response.data]

# 3. Assign the vectors as a new column in the DataFrame
headlines_df["embedding_vector"] = vector_embeddings

headlines_df
```

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	headline_text	embedding_vector	
0	Political leaders convene for summit on climat...	[-0.01142120361328125, -0.08221435546875, 0.15...	
1	New breakthrough in renewable energy technolog...	[-0.019256591796875, -0.037200927734375, -0.01...	
2	Global warming activists rally in major cities...	[0.11279296875, 0.06317138671875, 0.1616210937...	
3	Sports teams unite to raise awareness about cl...	[-0.036041259765625, -0.0411376953125, 0.09210...	
4	Government announces new policy to combat clim...	[-0.01354217529296875, -0.01506805419921875, 0...	
...	...	...	
118	Climate change documentary wins prestigious fi...	[0.0263824462890625, 0.0302581787109375, -0.05...	
119	Political candidates make final push in campai...	[0.110107421875, -0.005535125732421875, 0.0336...	
120	Sports team celebrates victory with parade thr...	[0.13330078125, -0.055328369140625, -0.0079040...	
121	Tech conference addresses cybersecurity threat...	[-0.00960540771484375, -0.0094451904296875, 0....	
122	Global warming impact on indigenous communitie...	[0.0958251953125, 0.029937744140625, 0.0987548...	

123 rows × 2 columns

Next steps:

[Generate code with headlines_df](#)[New interactive sheet](#)

Use `text-embedding-3-small` text embedding model to generate the embedding vectors with 256 dimensions of the headlines.

Start coding or [generate](#) with AI.

Extract the vector embeddings from the `response`

```
user_prompt = "global warming and climate change"
```

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I used the embedding model to generate a 256-dimensional embedding vector related to the phrase: `"global warming and climate change"`. then extract the embedding vector from the openAI response and store it into a variable called `search_vector`

Start coding or [generate](#) with AI.

```
response = client.embeddings.create(
    input = [user_prompt],
    model = "text-embedding-3-small",
    dimensions = 256
)
user_prompt_vvector = [d.embedding for d in response.data]
print(user_prompt_vvector[:10])
```

[0.0173492431640625, -0.058990478515625, 0.1474609375, 0.006805419921875, -0.06451416015625, 0.0229949951171875, 0.042236328125, -0.09

I created a dataframe containing the news headlines and their corresponding 256-dimensional vectors. My dataframe had two columns: `vectors` and `headlines`.

```
# Create the dataframe with the specified column names
df = pd.DataFrame({
    "vectors": vector_embeddings,
    "headlines": headlines
})

# Preview the dataframe
df.head()
```

	vectors	headlines
0	[-0.01142120361328125, -0.08221435546875, 0.15...	Political leaders convene for summit on climat...
1	[-0.019256591796875, -0.037200927734375, -0.01...	New breakthrough in renewable energy technolog...
2	[0.11279296875, 0.06317138671875, 0.1616210937...	Global warming activists rally in major cities...
3	[-0.036041259765625, -0.0411376953125, 0.09210...	Sports teams unite to raise awareness about cl...
4	[-0.01004211329290013, -0.01506805419921875, 0...	Government announces new policy to combat clim...

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Next steps:

[Generate code with df](#)

[New interactive sheet](#)

I used the `cosine_similarity` measure to calculate the similarity between the `search_phrase_vector` and each of the embedding vectors of the headlines.

My code was written in a way that includes a new column in the dataframe called `similarity_score`.

```
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity

# 1. Convert the list of vectors from the DataFrame into a clean 2D matrix
matrix_embeddings = np.array(df["vectors"].tolist())

# 2. Map it directly without extra brackets since it's already 2D
search_matrix = np.array(user_prompt_vvector)

# 3. Calculate similarity
df["similarity_score"] = cosine_similarity(matrix_embeddings, search_matrix).flatten()

# Preview the sorted results to check accuracy
df.sort_values(by="similarity_score", ascending=False).head()
```

	vectors	headlines	similarity_score	
98	[0.005901336669921875, 0.034942626953125, 0.12...	Global warming effects observed in changing we...	0.619019	
42	[0.0121307373046875, -0.00653839111328125, 0.1...	Global warming impacts discussed at internatio...	0.607141	
58	[0.0780029296875, 0.014617919921875, 0.0787353...	Global warming awareness campaign gains momentum	0.572996	
86	[-0.01171112060546875, 0.005260467529296875, 0...	Climate change task force formed to address ur...	0.558026	
22	[0.03424072265625, 0.0241546630859375, 0.14709...	Climate change impacts highlighted in new scie...	0.551199	

I sorted the dataframe by the `similarity_score` column and found 10 headlines that are related to the search phrase.

```
# Sort (Disk: 20.49 GB/107.72 GB similarity first) and slice the top 10 rows
top_10_headlines = list(df.sort_values(by="similarity_score", ascending=False).head(10).headlines)

# Display the results
top_10_headlines
```



```
['Global warming effects observed in changing weather patterns',  
'Global warming impacts discussed at international conference',  
'Global warming awareness campaign gains momentum',  
'Climate change task force formed to address urgent issues',  
'Climate change impacts highlighted in new scientific report',  
'Climate change summit results in historic agreement',  
'Global warming effects felt in Arctic region, scientists say',  
'Global warming solutions proposed at climate change forum',  
'Climate change summit attracts attention from around the world',  
'Global warming threatens food security, scientists caution']
```

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